List of PhD Scholars

S. No.	PhD Scholar Name/University	PhD thesis Topic Name	Registration Number	Publication
1	Azmat Hayat Khan Department of Meteorology, COMSATS Institute of Information Technology, Islamabad	"Investigation of Mechanisms to Detect Recurrence of Droughts in South Asia with Special Reference to Pakistan"	CIIT/FA 10-PME-003/ISB	Hanif M, Khan AH, &**Adnan S**. (2013). Latitudinal precipitation characteristics and trends in Pakistan. *Journal of Hydrology* 492: 266-272. **(IF: 2.693)**
2	Sajjad Haider Department of Meteorology, COMSATS University Islamabad, Islamabad	Assessment of Observed and Projected Aridity and Climatic Extremes over Agro-climatically Diversified Region of Pakistan under Climate Change Scenarios	CIIT/SP13-PME-004/ISB	Haider, S., & **Adnan, S.**(2014). Classification and assessment of aridity over Pakistan provinces (1960-2009). *International Journal of Environment*, *3*(4), 24-35
3	Ghazala Qaiser Department of Meteorology, COMSATS University Islamabad, Islamabad	Development and Application of Composite Drought Index for Drought Assessment in Pakistan	CIIT/FA-13 PME-001/ISB	Qaiser, G., Tariq, S., **Adnan, S**., & Latif, M. (2021).Evaluation of a composite drought index to identify seasonal drought and its associated atmospheric dynamics in Northern Punjab, Pakistan. *Journal of Arid Environments*, *185*, 104332. **(IF: 2.759).**
4	Imran Ahmed Durrani Balochistan University of Information Technology, Engineering and Management Sciences Quetta	Assessment of Climate Change Impacts on the Groundwater of Quetta Valley	BUITEMS/EMP/F-07/72-PhD	• Durrani, I. H., Adnan, S.,& Aftab, S. M. (2021). Daily Climate Extremes of Temperature and Precipitation Over Quetta Valley, Pakistan During 1961-2019. *International Journal of Environment*, *10*(1), 20-48. • Durrani, I. H., **Adnan, S**., & Aftab, S. M. (2018). Historical and Future Climatological Drought Projections Over Quetta Valley, Balochistan, Pakistan. In *IOP Conference Series: Materials Science and Engineering* 414(1): 012043. IOP Publishing. **(IF: 7.328)** • Durrani, I. H., **Adnan, S.,** Ahmad, M., Khair, S. M., & Kakar, E. (2018). Observed long-term climatic variability and its impacts on the ground water level of Quetta alluvial. *Iranian Journal of Science and Technology, Transactions A: Science*, *42*(2), 589-600.**(IF: 0.692)**
5	Arbab Siddique Department of Biosciences COMSATS University Islamabad, Islamabad	Incidence of dengue and its associated microbiome during pregnancy	CIIT/FA 16-PBS-003/ISB	Saddique, A., **Adnan, S.,** Bokhari, H., Azam, A., Rana, M. S., Khan, M. M. ...& Sharif, S. (2021). Prevalence and Associated Risk Factor of COVID-19 and Impacts of Meteorological and Social Variables on Its Propagation in Punjab, Pakistan. *Earth Systems and Environment*, **1-14. (IF: 7.2).**

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List of MS Scholars

S. No.	MS Scholar Name/Registration Number/University	MS thesis Topic Name	Publication
1	Syed Waseem Iqbal CIIT/SP18-RME-001 Department of Meteorology, COMSATS University Islamabad, Islamabad	Performance evaluation and comparison of observed and reanalysis gridded precipitation datasets over Pakistan	Iqbal, S. W., Latif, M., Ahmed, R., **Adnan, S.,** Umar, M., Ahmad, B. ...& Syed, J. H. (2022). Performance evaluation and comparison of observed and reanalysis gridded precipitation datasets over Pakistan. *Theoretical and Applied Climatology*, 1-24.**(IF: 3.409).**
2	Maida Ashraf CIIT/FA17-RRG-005/ISB Department of Meteorology, COMSATS University Islamabad, Islamabad	Satellite based impact assessment of temperature and rainfall variability on drought indices in Southern Pakistan.	Ashraf, M., Ullah, K., &**Adnan, S.** (2022). Satellite based impact assessment of temperature and rainfall variability on drought indices in Southern Pakistan. *International Journal of Applied Earth Observation and Geoinformation*, *108*, 102726. **(IF: 7.672).**
3	Khadija Ashraf Department of Meteorology, COMSATS University Islamabad, Islamabad	Spatio-temporal assessment and climatology of atmospheric organic carbon over Pakistan	Hashmi, M. Z., Ashraf, K., **Adnan, S.**, Wang, S., Pongpiachan, S., Su, X., & Ahmed, Z. (2020). Spatio-temporal assessment and climatology of atmospheric organic carbon over Pakistan. *Arabian Journal of Geosciences*, *13*(15), 1-13.**(IF: 1.827).**
4	Asma Akhtar Department of Environment Sciences International Islamic University Islamabad	Analyzing the occurrence of floods and droughts in connection with climate change in Punjab province, Pakistan.	Ali, S. M., Khalid, B., Akhter, A., Islam, A., &**Adnan, S.** (2020). Analyzing the occurrence of floods and droughts in connection with climate change in Punjab province, Pakistan. *Natural Hazards*, *103*(2), 2533-2559.**(IF: 3.102).**
5	Noor-Ul-Ain CIIT/FA16-RME-002/ISB Department of Meteorology, COMSATS University Islamabad, Islamabad	Investigation of seasonal droughts and related large-scale atmospheric dynamics over the Potwar Plateau of Pakistan.	Ain, N.U., Latif, M., Ullah, K., **Adnan, S.**, Ahmed, R., Umar, M. and Azam, M., (2020. Investigation of seasonal droughts and related large-scale atmospheric dynamics over the Potwar Plateau of Pakistan. *Theoretical and Applied Climatology*, pp.1-21.**(IF: 3.179).**
6	Fayyaz Ahmed Department of Meteorology, COMSATS University Islamabad, Islamabad	Impact of jet stream and associated mechanisms on winter precipitation in Pakistan	Ahmed, F., **Adnan, S.,** & Latif, M. (2020). Impact of jet stream and associated mechanisms on winter precipitation in Pakistan. *Meteorology and Atmospheric Physics*, *132*(2), 225-238. **(IF: 2.065)**
7	Muhammad Roman Khan Institute of Geo-Information & Earth Observation, PMAS Arid Agriculture University Rawalpindi	Assessing meteorological and agricultural drought in Chitral Kabul river basin using multiple drought indices.	Baig, M. H. A., Abid, M., Khan, M. R., Jiao, W., Amin, M., &**Adnan, S.** (2020). Assessing meteorological and agricultural drought in Chitral Kabul river basin using multiple drought indices. *Remote Sensing*, *12*(9), 1417. **(IF: 4.848).**
8	Fazale Rehman 00000274254 School of Civil and Environmental Engineering (SCEE), National University of Sciences and Technology (NUST), Islamabad	Identification and Assessment of Agrometeorological Droughts, Potohar Plateau of Pakistan	Identification and Assessment of Agrometeorological Droughts, Potohar Plateau of Pakistan (Publication in process)
9	Saad Nisar 00000204735 School of Civil and Environmental Engineering	Investigation of Meteorological & Agricultural drought in rain-fed areas of	Investigation of Meteorological & Agricultural drought in rain-fed areas of Sindh/ South-Eastern Sindh (Publication in Process)

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	(SCEE), National University of Sciences and Technology (NUST), Islamabad	Sindh/ South-Eastern Sindh	
10	Syed Usama Imtiaz 00000206076 School of Civil and Environmental Engineering (SCEE), National University of Sciences and Technology (NUST), Islamabad	Estimation of Chenab River Water Losses Using Mass Balance Approach.	
11	Eng. Muhammad Zaheer 00000276945 School of Civil and Environmental Engineering (SCEE), National University of Sciences and Technology (NUST), Islamabad	Identification and assessment of seasonal Trends of meteorological and agricultural droughts – a case study of thal region, Punjab, Pakistan	Identification and assessment of seasonal Trends of meteorological and agricultural Droughts – a case study of thal region, Punjab, Pakistan(Publication in Process)
12	Asma Hafeez FA19-RRG-002 Department of Meteorology, COMSATS University Islamabad, Islamabad	Effects of Changes in Land Surface temperature and Precipitation Patterns on flood occurrence in the Upper Indus Basin.	
13	Abdul Basit F2016123004 School of Governance and Society (SGS), University of Management and Technology	Policy Analysis of National Climate Change Policy of Pakistan - 2012	
14	Sajjad Haider CIIT/FA11-RRG-007/ISB Department of Meteorology, COMSATS Institute of information Technology, Islamabad	Spatio-Temporal Analysis of Aridity over Pakistan for the period 1960-2011	
15	Maryam Pervaiz 18-arid-1490 Department of Zoology, PMAS Arid Agriculture University Rawalpindi	Iron deficiency anemia in female students of PMAS Arid Agriculture University Rawalpindi	
16	Habiba Khalid CIIT/SP13-REM-003 Department of Energy Management, COMSATS Institute of Information Technology Islamabad	Empirical Modeling of Global Solar Radiation over Southern Areas of Pakistan	Khalid, H., & Zakaria, M. (2016). Estimation of solar radiation in southern areas of Pakistan using radiation models. *Journal of Renewable and Sustainable Energy*, *8*(4).
17	Hamna Rasool 01-161132-044 Department of Earth and Environment Sciences, Bahria University, Islamabad	Investigation of Smog Events Over Lahore city from the year 2012-2017	
18	Aizaz Ahmed 00000328077 Institute of Geographical Information Systems (IGIS) National University of Science and Technology (NUST), Islamabad	Glacier Mass balance and Velocity analysis using InSAR and Offset Tracking Technique: A case study of Batura, Passu, Gulkin and Gulmit Glaciers, Hunza valley (in Progress)	
19	Hammad Javid	Improving Flood Forecasts in Pakistan Using Global Flood Awareness System	

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	School of Geography and the Environment, University of Oxford, UK		
20	Syeda Areeba Gillani 14-Arid-5001 Faculty of Agricultural Engineering &Technology PMAS Arid Agriculture University Rawalpindi	Drought monitoring of Balochistan using Geospatial techniques	
21	Syed Zeeshan Haider CIIT/FA16-RRG-003/ISB Department of Meteorology COMSATS University Islamabad, Islamabad	Coverability of Aerosols and Monsoon Rainfall over Sialkot, Lahore and Faisalabad, Pakistan	
22	Syeda Saba Fatima FA19-RRG-016 Department of Meteorology COMSATS University Islamabad, Islamabad	Assessment of Heatwaves Impacts on Human Health in Sindh and Punjab Provinces using Remote Sensing Techniques	

List of BS Scholars

S. No.	BS Scholar Name/Registration Number/University	BS Project Title	Publication
1	Alishba Arshad BS.Env- 2018-003 Department of Environmental Sciences, Fatima Jinnah Women University	Climate Models evaluation for 21st Century	
2	Mahnoor Ajmal, Mahnur Mughal, Maryam Asif, Maryum Tufail, Department of Environmental Sciences, International Islamic University Islamabad	Consequences of climatic parameters on Pollen Count and its effects on human health	

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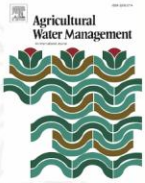

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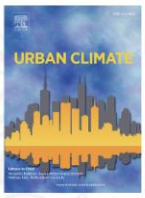

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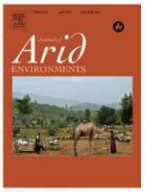

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National Drought Monitoring Centre
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Islamabad, Pakistan

Islamabad, 14th May, 2022

Subject: Technical Report on catastrophe due to heatwave instead of drought in the Cholistan, Punjab

Summary

The catastrophe caused by the severe heatwave in Cholistan impacted the water reservoirs, vegetation and livestock badly. Due to abnormal increases in daily maximum temperature, heat stress situations arose in the Cholistan region. . The high-intensity major heatwave persist for 41 days (11th March to 19th April 2022) followed by a short heatwave of 6 days (27th April to 2nd May 2022) of relatively low intensity. Although the region received a sufficient amount of rainfall during the winter season, however, the severe heatwave is now alarming the onset of drought in the coming days in the Cholistan region due to water depletion in ponds and reservoirs as well as soil moisture availability for vegetation.

(Note: This report is prepared under the special directive of Mehr Sahibzad Khan, Director General, PMD, Islamabad.)

1. Introduction

The Cholistan comprises three districts Bahawalnagar, Bahawalpur, and Rahim Yar Khan. The total area of Cholistan is 66, 55,360 acres. The largest area of Cholistan is present in Bahawalpur which is 40, 28,217 acres. The length of Cholistan is 480 km and the width ranges from 32 to 192 km.

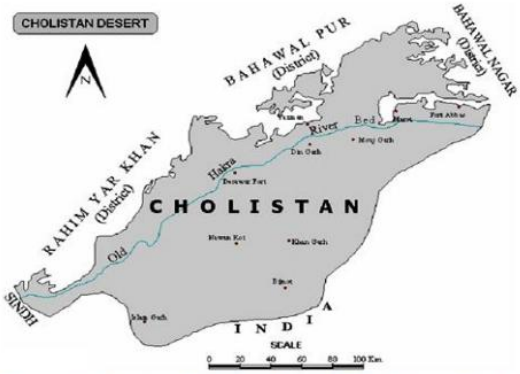

Figure 1 Geographical location of Cholistan, Punjab

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Pakistan Meteorological Department
Islamabad

3. Heatwave situation

The daily temperature remains low during winter and it gradually increases during transition months i.e., March and April. However, a sharp increase in daily temperature was observed over most parts of the country, especially the Cholistan region. The first high-intensity major heatwave with duration of 41 days was observed from 11th March to 19th April 2022 in which daily maximum temperature remained consistently above 5°C to 11.5°C than normal over all the districts of Cholistan (Fig 5). Another heatwave of relatively low intensity was observed from 27th April to 2nd May with a duration of 6 days where the daily maximum temperature was above 3.8 to 6.8°C than normal (Fig 5).

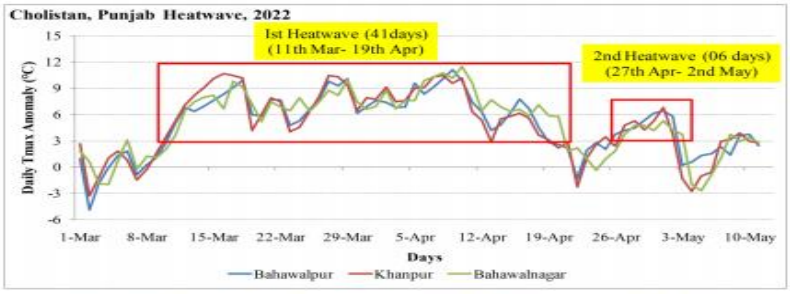


Figure 5 Heatwave episodes experienced in the Cholistan region, Punjab.

Diurnal variation in body temperature, ranging from about 37.5 °C from 10 a.m. to 6 p.m., and falling to about 36.4 °C from 2 a.m. to 6 a.m. ***Cheng et. al. (2014)*** identified the effects of Diurnal Variation of Temperature (DVT) on human health. The DVT is significantly associated with mortality and morbidity, particularly for cardiovascular and respiratory diseases.

All animals have a thermal comfort zone, which is a range of ambient environmental temperatures that are beneficial to physiological functions (FAO, 1986). During the day, livestock keeps a body temperature within a range of ±0.5 C (Henry et al., 2012). When temperature increases more than the upper critical temperature of the range (varies by species

type), the animals begin to suffer heat stress (FAO, 1986). Warm and humid conditions that cause heat stress can affect livestock mortality. Howden et al. (2008) reported that increases in temperature between 1 and 5 °C might induce high mortality in grazing cattle. The daily diurnal temperature variation recorded in Cholistan was 14.9°C to 22.7°C during the heatwave period (Fig 6).

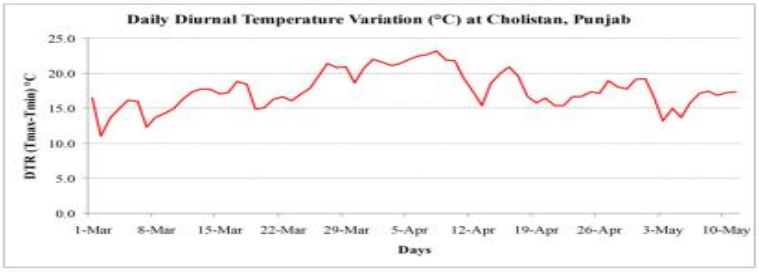


Figure 6 Daily diurnal temperature variations in the Cholistan, Punjab

4. Conclusion

Based on the above results, it is concluded that severe heatwave impacted the vegetation, water resources and most importantly the livestock in the Cholistan region instead of drought. The severe heat waves trigger the evapotranspiration by 1.5mm/day to 3.5mm/day which impacted the soil moisture, water table/resources and vegetation in the region. The high temperature caused heat stress which affected livestock mortality. Furthermore, high variation in the daily diurnal temperature (≥ 5 °C) increased the probability of high mortality in grazing cattle. Finally, this heatwave has also enhanced the probability of the drought emergence in the Cholistan region.

(Dr. Shahzada Adnan)
Deputy Director
For Director General

Dengue Prevalence and its association with Meteorological Parameters

Dr. Shahzada Adnan
Meteorologist
Pakistan Meteorological Department

Dengue Life Cycle

Life Cycle :

10-12 Days

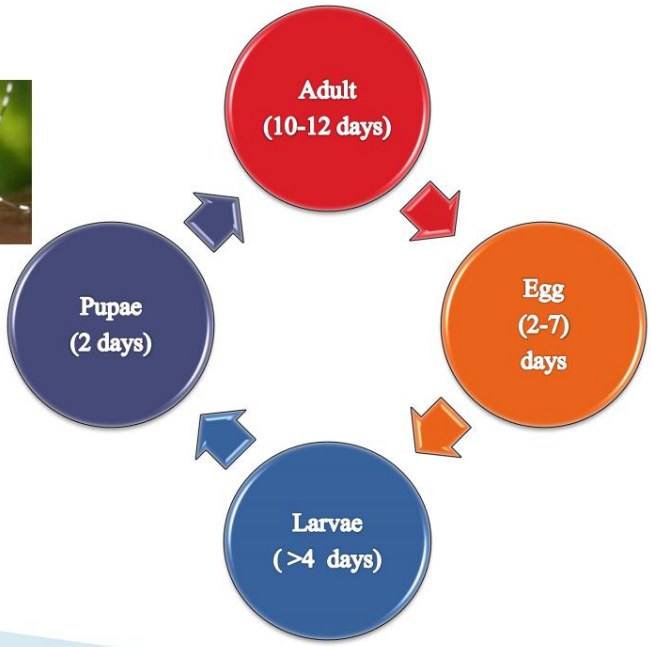
Egg survival period:

8 months Approx.

Active period for dengue attack :

(Two hour after sunrise) &

(Two hours before sunset)



Pakistan Meteorological Department
Climate Change Impact and Integration Cell (CIIC)
Meteorological Headquarters
Islamabad, Pakistan

Islamabad, 23rd September 2022

SUBJECT: Dengue Alert-I

Dengue fever has deepened its roots and impacted severely the health of people over the last ten years in the country. This phenomenon occurs especially during the post-monsoon season (20th September till 5th December) provided the conditions are favourable. The dengue triggers during those periods where the temperature and humidity threshold remains 26-29°C (3-5 weeks) and 60% respectively. Moreover, the rainfall remains ≥ 27mm with a maximum time lag period of 3 weeks. The active period for dengue attacks is two hours after sunrise and two hours before sunset. The breeding stops once the temperature falls below 16°C.

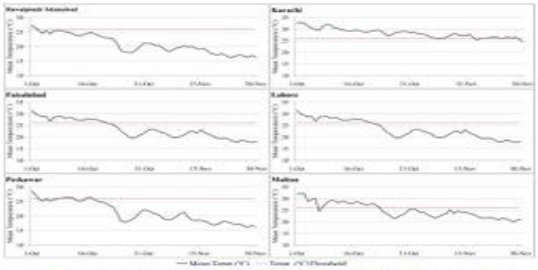


Figure 1 Temperature forecast and threshold of Dengue vulnerable districts during Oct-Nov 2022

2. Keeping in view the past data analysis, current and future climate outlook, the above-mentioned atmospheric variables are providing an ideal environment for a dengue outbreak. Based on the analysis, the environment has become conducive from mid of September 2022 for dengue onset and it is predicted that it may outbreak in October 2022, particularly in ten major cities of Pakistan i.e., Karachi, Lahore, Peshawar, Rawalpindi, Islamabad, Hyderabad, Faisalabad, Sialkot, Larkana and Multan as well as in flood affected areas of Pakistan.

3. It is advised to all stakeholders to take pre-emptive measures for the dengue outbreak in the districts. National Health agencies and Dengue control centres are advised to keep themselves updated on the PMD website http://www.pmd.gov.pk.

(Dr. Shahzada Adnan)
Meteorologist
For Director General

Pakistan Meteorological Department
Climate Change Impact and Integration Cell (CIIC)
Meteorological Headquarters
Islamabad, Pakistan

Islamabad, 27th September 2023

SUBJECT: Dengue Alert-I

Dengue fever has deepened its roots and impacted severely the health of people over the last ten years in the country. This phenomenon occurs especially during the post-monsoon season (20th September till 5th December) provided the conditions are favourable. The dengue triggers during those periods where the temperature and humidity threshold remains 26-29°C (3-5 weeks) and 60% respectively. Moreover, the rainfall remains ≥ 27mm with a maximum time lag period of 3 weeks. The active period for dengue attacks is two hours after sunrise and two hours before sunset. The breeding stops once the temperature falls below 16°C.

2. Keeping in view the past data analysis, current and future climate outlook especially the late retreat of monsoon, the above-mentioned atmospheric variables are providing an ideal environment for a dengue outbreak. Based on the analysis, the environment has become conducive from mid of September 2023 for dengue onset and it is predicted that it may outbreak in October 2023, particularly in ten major cities of Pakistan i.e., Karachi, Lahore, Peshawar, Rawalpindi, Islamabad, Hyderabad, Faisalabad, Sialkot, Larkana and Multan as well as in post monsoon rainfall affected areas of Pakistan.

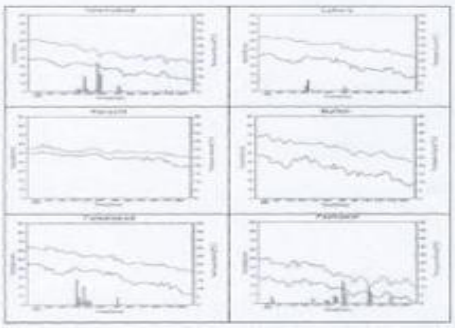


Figure 1 Precipitation and Temperature forecast of Dengue vulnerable districts during Oct-Nov 2023.

3. It is advised to all stakeholders to take pre-emptive measures for the dengue outbreak in the districts. National Health agencies and Dengue control centres are advised to keep themselves updated on the PMD website http://www.pmd.gov.pk

(Dr. Shahzada Adnan)
Deputy Director
For Director General

Government of Pakistan
Pakistan Meteorological Department
National Drought Monitoring Centre
Meteorological Headquarters
Islamabad, Pakistan

Islamabad, 14th May, 2022

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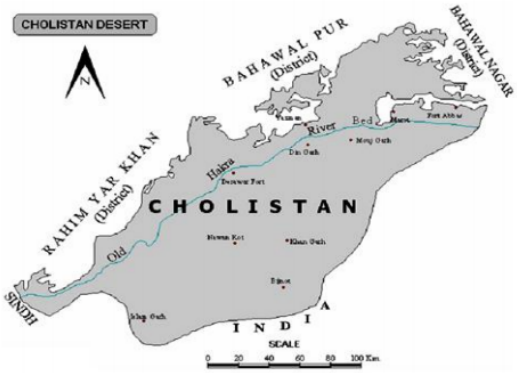

Figure 1 Geographical location of Cholistan, Punjab

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